



MRK HOTELS

Review 04 — Laundry Cloth-Type Registry

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MRK Hotels — Review Batch 04 · Laundry Cloth-Type Registry & Auto-Pricing (Problem & Solution)

Module: Laundry — **Status:** Resolved — **5 September 2026**

1. The Problem

Laundry line items were built from **free-typed names with hand-entered unit prices**. There was no price list and no inventory of the cloth/linen types a hotel actually launders.

What this meant in practice:

- Prices drifted: the same "bed sheet" could be charged differently by different attendants, or across days.
- No hotel had a manageable, reviewable laundry price list (wash / iron / dry-clean).
- Data was inconsistent — typo'd item names made reporting and charges unreliable.

2. Root Cause

- `laundry_order_items` had only free-text `item_name` + numeric `unit_price`, with **no link** to any registered product.
- There was **no model, controller or routes** for a cloth-type catalogue.
- The order endpoints simply trusted whatever `unit_price` the client sent; nothing locked the price to a list.

3. The Solution

3.1 Backend — a first-class cloth-type registry

- New table `cloth_types` (via migration):

Column	Notes
<code>cloth_type_id</code>	UUID primary key
<code>tenant_id</code>	FK → hotels (tenant scoped)
<code>name</code>	Display name (e.g. "Bed sheet")
<code>wash_price</code> , <code>iron_price</code> , <code>dry_clean_price</code>	Price per service
<code>is_active</code>	Soft enable/disable for the item picker

Column	Notes
timestamps	created_at / updated_at

- `laundry_order_items.cloth_type_id` added (nullable FK, `nullOnDelete`) so existing free-form items keep working.
- New `App\Models\Laundry\ClothType` model with a `priceFor(service)` helper (wash / iron / dry_clean → the matching price column).
- New `ClothTypeController` with full CRUD (`index` , `show` , `store` , `update` , `destroy`), tenant-scoped and audit-logged like the rest of the API.
- New routes in `api.php` under `laundry-cloth-types` :
- `GET /laundry-cloth-types` — level 40
- `POST/PUT/DELETE` — level 40 + operate permission

3.2 Backend — prices locked to the registry

- `LaundryOrderController` now validates `items.*.cloth_type_id` (`exists:cloth_types,cloth_type_id`).
- New `resolveItemUnitPrice($item, $service)` :
- When the item selects a registered cloth type, the **registry price for the order's service wins** — the client's `unit_price` is ignored.
- Free-form items (no cloth type) keep whatever price was sent.
- `store` and `update` compute every item's `subtotal` , `items_count` and `total_charge` from these resolved prices, so the estimate can never drift from the published price list.

3.3 Frontend — manage the registry

- New `src/pages/laundry/ClothTypesPage.vue` : searchable table, add/edit modal, delete; gated by `canManage` (level 40 + operate).
- New route `laundry/cloth-types` mounted under the Store layout.
- New `clothTypeApi` client (`index` / `store` / `update` / `destroy`).
- `18n` keys added to **both** `en.json` and `sw.json` (`clothTypes.*` + `laundry.*`).

3.4 Frontend — use the registry in the order form

- The item picker in `LaundryListPage.vue` is now a **SearchableSelect over active cloth types**.
- Selecting a cloth type auto-fills:
 - `item_name` = the cloth type's name,
 - `unit_price` = the registry price **for the currently selected service**.
- The price input becomes **read-only** once a cloth type is chosen.
- Switching the order's service re-prices every registry-backed line automatically (a `watch` on `form.service`).
- When a hotel has **no cloth types** registered yet, the form falls back to the legacy free-text + manual price input, so nothing breaks for hotels that haven't populated the list.

4. Files Changed

Area	File	Change
Backend	database/migrations/ 2026_09_05_000004_create_cloth_types_table.php	New <code>cloth_types</code> table
Backend	database/migrations/ 2026_09_05_000005_add_cloth_type_id_to_laundry_order_items.php	<code>cloth_type_id</code> FK on items
Backend	app/Models/Laundry/ClothType.php	Model + <code>priceFor()</code>
Backend	app/Models/Laundry/LaundryOrderItem.php	<code>cloth_type_id</code> fillable + <code>clothType()</code> relation
Backend	app/Http/Controllers/Api/V1/Laundry/ClothTypeController.php	CRUD controller
Backend	app/Http/Controllers/Api/V1/Laundry/LaundryOrderController.php	Validation + <code>resolveItemUnitPrice()</code> auto-pricing
Backend	routes/api.php	<code>laundry-cloth-types</code> routes
Frontend	src/pages/laundry/ClothTypesPage.vue	Registry management page
Frontend	src/api/index.js	<code>clothTypeApi</code>
Frontend	src/router/index.js	<code>laundry/cloth-types</code> route
Frontend	src/pages/laundry/LaundryListPage.vue	Cloth-type item picker + auto price
Frontend	src/locales/en.json , sw.json	New keys (both languages)
Tests	tests/Feature/ClothTypeTest.php	Registry CRUD + auto-pricing tests

5. Verification Results

- New `ClothTypeTest` covers: CRUD round-trip, tenant isolation, and auto-pricing for **wash + iron** from the registry.
- Backend full suite — **569 passed / 2305 assertions** (incl. migration-phase policy check for the two new migrations).
- Frontend `vite build` — **clean**; Vitest **107/107 passed**.
- Migrations verified additive in `up()` per the repo's expansion-phase policy.

6. Result

Every hotel now has a real, manageable laundry price list. Selecting a cloth type produces the correct item name and service-specific price automatically, order totals are computed server-side from the published list, and hotels that haven't set up cloth types keep the original free-form flow.